

Q2 Report: Statistical Analysis of Diabetes Dataset

This report presents the statistical analysis performed on the Diabetes dataset, following the assignment requirements. The objectives include data cleaning, sampling analysis, percentile comparison, and bootstrap evaluation.

1. Data Preparation: Imputation of Missing Values

The dataset contains 768 observations. Several columns such as Glucose, BloodPressure, SkinThickness, Insulin, and BMI contained zero values, which are biologically impossible. These values were treated as missing.

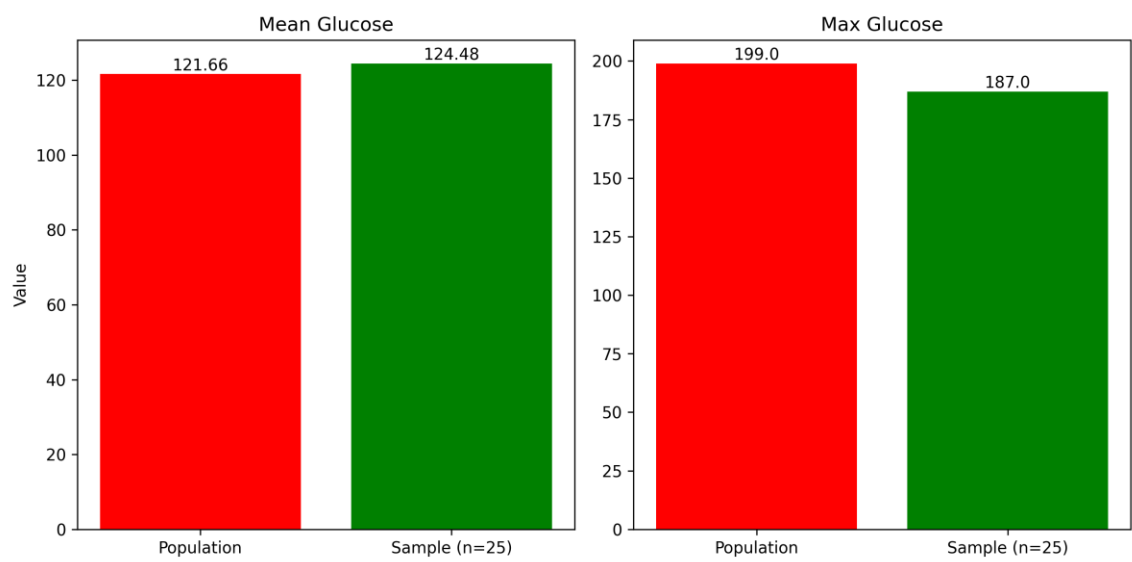
All zero values were replaced with NaN, and missing values were imputed using the median of each column.

2(a). Glucose: Sample vs Population

A random sample of 25 observations was taken using a fixed seed (101). The mean and maximum Glucose values were compared with the population values.

Findings: The sample statistics differed from the population, demonstrating that small samples can produce unstable estimates.

Figure 1: Add Glucose comparison chart here

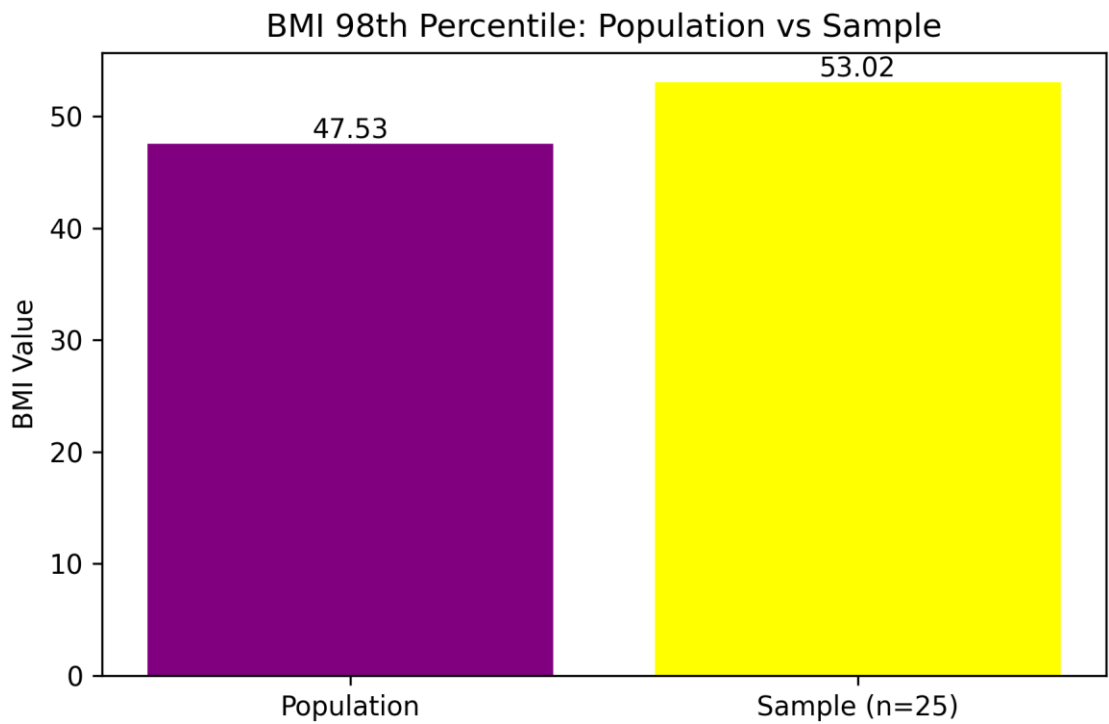


2(b). BMI 98th Percentile Comparison

The 98th percentile of BMI was calculated for both the sample and the population.

Findings: The sample percentile differed from the population, showing that extreme values are sensitive to sampling variability.

Figure 2: Add BMI percentile chart here

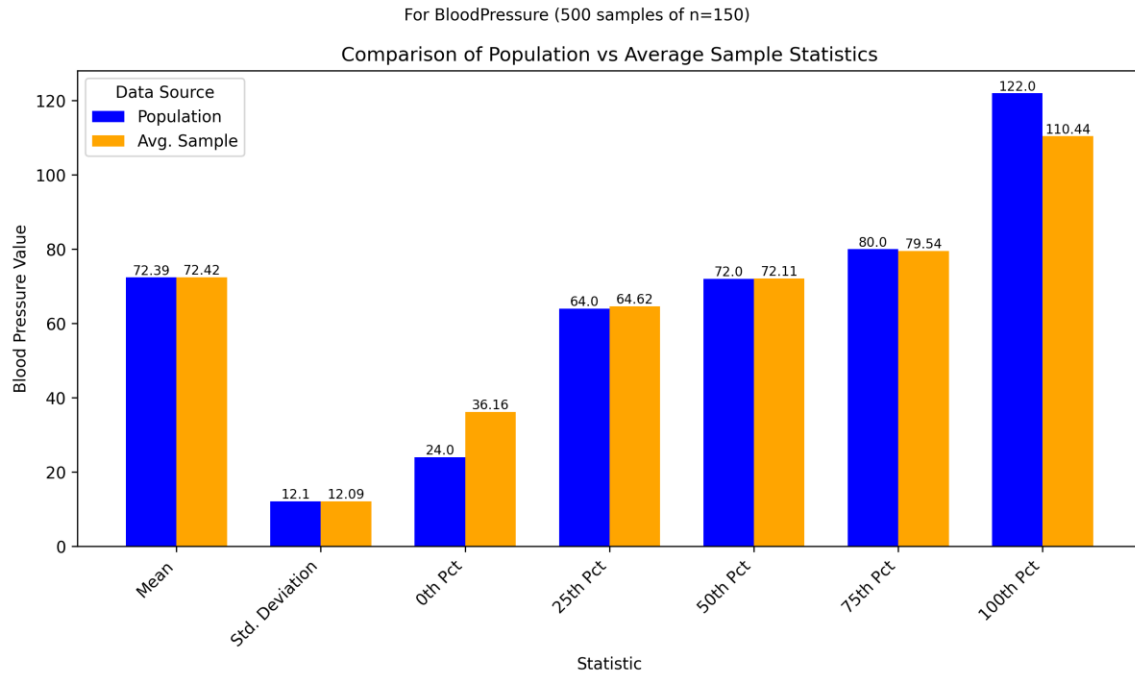


2(c). BloodPressure Bootstrap Analysis

Bootstrap sampling was performed using 500 samples of size 150 with replacement. The mean, standard deviation, and 98th percentile were computed and averaged.

Findings: The bootstrap results closely matched the population statistics, demonstrating that bootstrap sampling provides reliable estimates.

Figure 3: Add bootstrap comparison chart here



3. Conclusion

The analysis shows that small samples can produce high variability, while bootstrap sampling with larger sample sizes provides stable and accurate estimates of population statistics.